Experiment 6: Basic N-Gram Model for Text Generation in Python

1. Aim

To write a Python program to implement a basic N-gram (bigram) model for text generation.

2. Procedure

1. Import `random` and `defaultdict` in Python.
2. Define a sample training text and split it into words.
3. Build a bigram transition dictionary mapping each word to its following words.
4. Start with an initial word and repeatedly sample the next word from the bigram model.
5. Display the bigram dictionary and the generated text sequence.

3. Program

```
import random
from collections import defaultdict

random.seed(1)
words = "the cat sat on the mat the cat ate".split()
bigrams = defaultdict(list)

for w1, w2 in zip(words[:-1], words[1:]):
    bigrams[w1].append(w2)

curr, res = "the", ["the"]
for _ in range(5):
    curr = random.choice(bigrams[curr])
    res.append(curr)

print("Generated Text:", " ".join(res))
```

4. Output

```
Generated Text: the cat sat on the mat
```

5. Result

Thus, the Python program was successfully executed to implement a basic bigram model and generate text.